

Scalability & Future Plan

Date	01 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

The Credit Card Approval Prediction project is designed to provide accurate and efficient credit approval predictions. As the number of users and data grows, the system can be enhanced to improve performance, scalability, and security. This document outlines the current limitations, scalability strategies, and future enhancements planned for the project.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited dataset used for model training	May reduce prediction accuracy for diverse applicant profiles	High
2	Supports single-user prediction at a time	Not suitable for large-scale banking operations	Medium
3	Basic authentication and security features	Increased security measures are needed for production use	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on cloud infrastructure with load balancing for multiple concurrent users.
2	Data Storage	Uses a local dataset	Integrate with a scalable cloud database for efficient data management.
3	Performance	Predictions are generated using a locally deployed model	Optimize the model and use cloud-based inference services for faster predictions.
4	Security	Basic input validation	Implement secure authentication, encryption, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add real-time database integration and applicant record management	Next 3 Months	Improve data storage and management efficiency.
Phase 3	Develop a mobile-friendly application and REST API	Next 6 Months	Increase accessibility across different devices and platforms.
Phase 4	Enhance the model using larger datasets and advanced deep learning techniques	Next 12 Months	Improve prediction accuracy and overall system performance.